



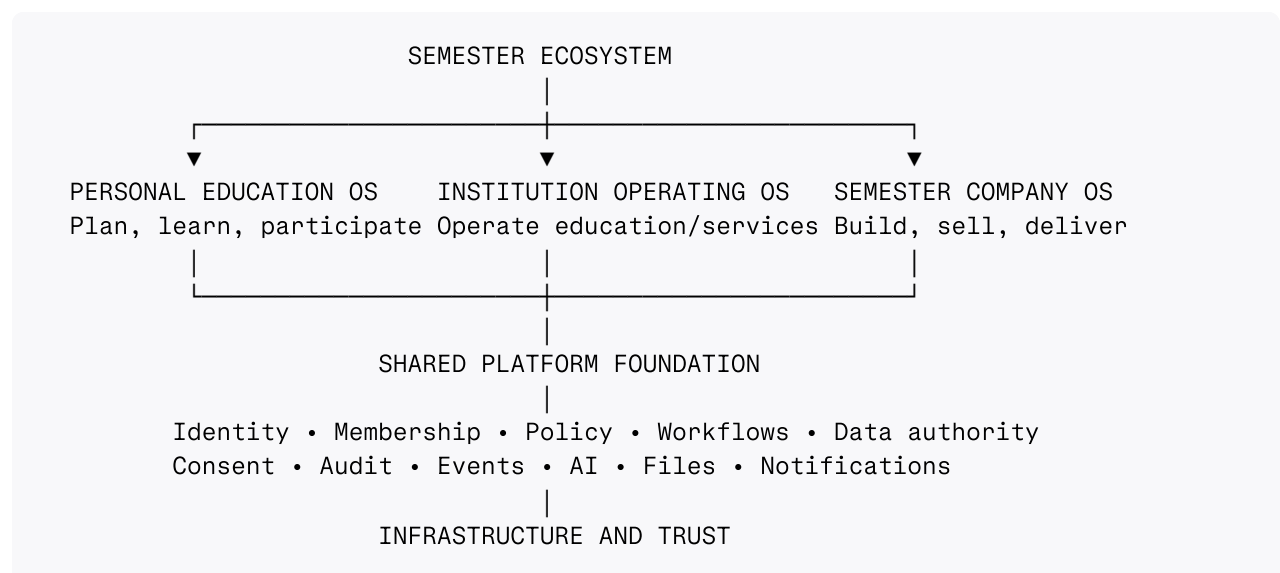
continue expanding illustrating and depicting the entire connected semester ecosystems platform

Semester's connected ecosystem is the entire loop from discovering the company to learning, teaching, running an institution, operating Semester itself, and continuing into employment and lifelong education. The expansion below shows not just what each system contains, but how people, records, decisions, money, and operational work move between them.

This is a proposed target architecture extending your mainframe specification—not a live audit or a claim that every capability is already built. Your document establishes the shared identity, native domain engines, data foundation, AI layer, control planes, migration boundary, and trust foundation. ^[1]

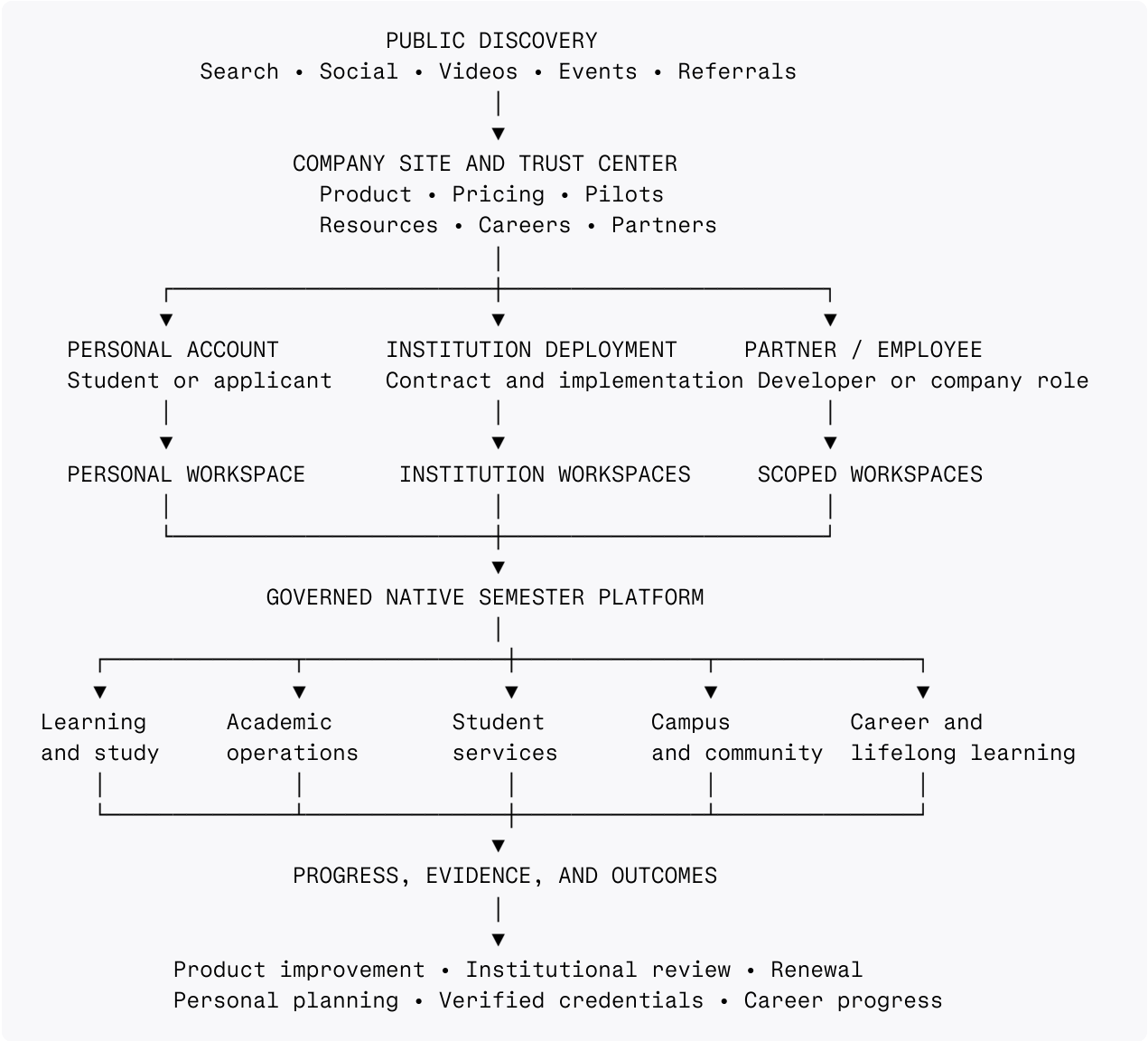
1. Connected ecosystem map

Think of Semester as three connected operating environments over one governed platform:



The environments are connected, but their access boundaries remain distinct. A student's personal notes, an institution's official academic records, Semester's employee records, and a developer's sandbox must not become one universally accessible dataset.

The ecosystem around Semester



Your mainframe document connects public acquisition, education applications, company operations, institutional administration, developer governance, and lifecycle management in this same overall structure. ^[1]

Four connection types

Use these distinctions when depicting or implementing the ecosystem:

Connection	What moves	Example	What must not happen
Experience connection	Navigation and approved context	Student opens an assignment from Today	Navigation must not grant access
Workflow connection	Requests, approvals, tasks, and receipts	Advisor refers a student to tutoring	A referral must not disclose unnecessary case details
Data connection	Authorized records or projections	Enrollment updates a course roster	A copied view must not become a conflicting official record

Connection	What moves	Example	What must not happen
Operational connection	Health, configuration, evidence, and support work	Failed integration creates an operations task	Platform staff must not automatically receive all customer data

The proposed architecture should make each connection explicit rather than using a vague “everything integrates” label.

2. Education journey network

The learner’s journey provides the organizing spine of Semester. Each stage activates different native systems and different human responsibilities.

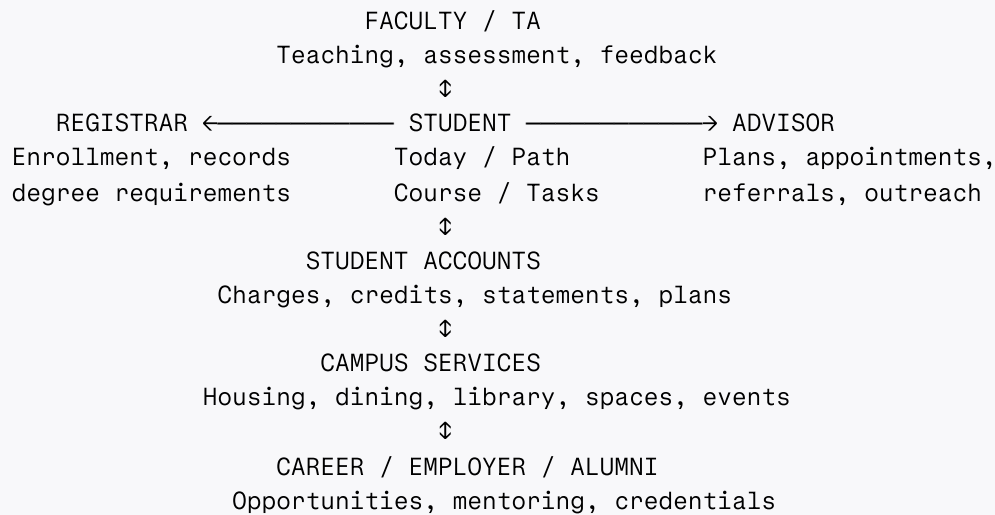


CONTINUE

Alumni • Further education • New credentials • Lifelong planning

These product areas extend the student, applicant, academic, campus, career, and lifecycle domains in your specification. Official graduation, transcript, and credential actions must remain governed institutional workflows.^[1]

The student-centered connection map



Additional scoped connections:

Student ↔ Guardian: explicit sharing grant

Student ↔ Community: membership and moderated interaction

Student ↔ AI: permitted assistance, sources, and proposed next actions

Each participant sees a purpose-built view. The student sees their next action; the advisor sees assigned support work; the registrar sees enrollment exceptions; finance sees account workflows. They do not all receive the same unrestricted “student profile.”

Example: registration to learning

One registration request can connect several systems without duplicating their responsibilities:

1. Student selects a planned section.
2. Registration engine checks eligibility and authoritative source state.
3. A prerequisite issue produces an override request.
4. The authorized reviewer receives an approval task.
5. Reviewer approves or denies under institution policy.
6. Authoritative enrollment system confirms the final result.
7. Enrollment event updates:
 - Student schedule
 - Academic plan
 - Faculty roster
 - Course access
 - Relevant operational queue

8. Student receives confirmation.
9. Any applicable financial effect enters a separate validated workflow.

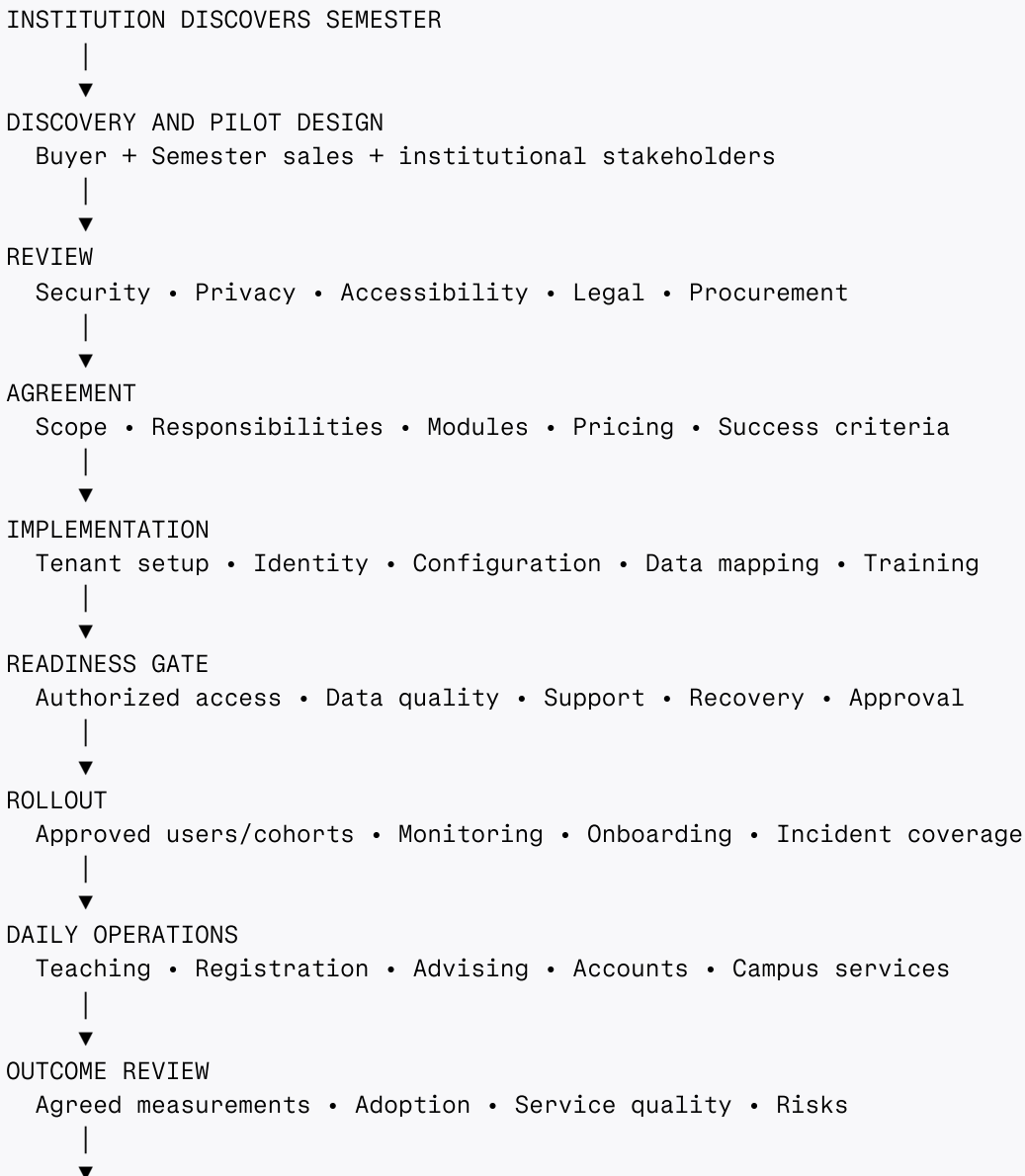
A planning choice is not enrollment. An approved override is not necessarily enrollment. A payment is not necessarily permission to enroll. These are separate domain facts connected by policy-controlled workflows.

Your document already specifies registration validation, authoritative confirmation, audit records, schedule/roster projections, and approved handoffs while another SIS remains authoritative.^[1]

3. Institution and company network

Semester must connect the customer lifecycle with the institutional operating lifecycle—not simply connect sales to an invoice.

Institution deployment lifecycle



RENEW / EXPAND / EXIT

Additional modules or campuses / renewal / governed portability

This follows the marketing-to-renewal operating loop in your mainframe document, including implementation, trust review, support, outcome evidence, and offboarding. ^[1]

Institutional departments stay distinct

Department	Native operating responsibilities	Connected workflows
Academic affairs	Programs, curriculum, academic policy	Registrar, faculty, degree planning
Registrar	Terms, sections, enrollment, records, graduation	Students, faculty, academic affairs
Teaching teams	Course delivery, assessment, feedback	Students, registrar, accessibility support
Student success	Advising, appointments, referrals, outreach	Students, academic teams, service offices
Student accounts	Charges, credits, payment plans, reconciliation	Students, approved aid/registration handoffs
Campus services	Housing, dining, library, bookings, service cases	Students, eligibility, service administration
Career and alumni	Opportunities, employers, mentoring, outcomes	Students, graduates, employers
IT and identity	SSO, provisioning, integrations, technical access	All authorized workspaces
Governance and assurance	Privacy, retention, accessibility, controls, risk	Institution administration and Semester trust teams
Institutional leadership	Approved aggregate outcomes, decisions, deployment status	Department leaders and customer-success review

These are operating views over shared platform capabilities, not ten independent applications.

Semester's company loop

GROWTH

Content, campaigns, referrals, events



REVENUE

Leads, accounts, opportunities, quotes, agreements



IMPLEMENTATION

Tenant activation, integrations, migration, training



CUSTOMER SUCCESS

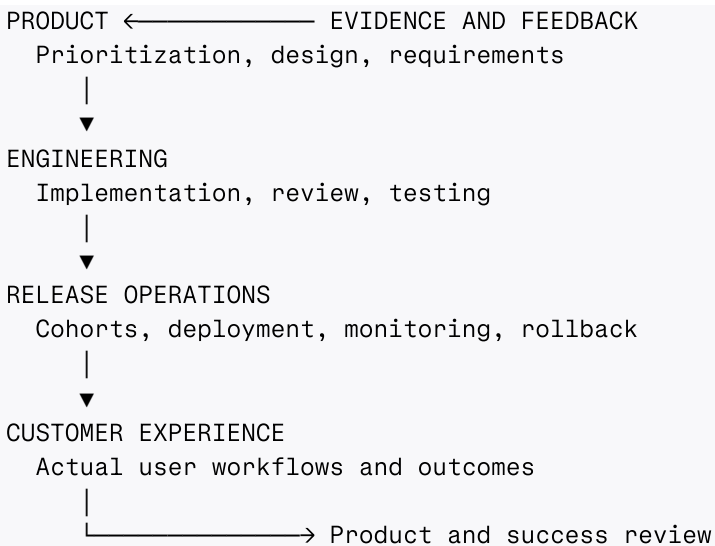
Adoption, agreed outcomes, reviews, renewal, expansion



SUPPORT

Cases, escalation, approved access





Finance, legal, people operations, security, privacy, accessibility, and AI governance support this loop through their own restricted workspaces.

Your specification includes these company functions and explicitly separates Semester employee capabilities from access to institution records.^[1]

Support versus engineering visibility

A proposed failed registration-support case illustrates that boundary:

```
Student reports missing confirmation
→ Support sees the case and minimum necessary context
→ Integration operator sees sync state and external reference
→ Registrar sees the permitted enrollment record
→ Engineer sees sanitized error telemetry
→ Security team receives a case only if a security concern is identified
```

“Connected” means the work can be routed to the right owner. It does not mean every team receives all underlying content.

4. Data, events, and intelligence

The ecosystem needs a record ownership map and an event network—not just a list of screens.

Record ownership map

```
IDENTITY DOMAIN
  Account → memberships → authorized workspace contexts

ACADEMIC DOMAIN
  Program → requirements → course → section → enrollment → record

LEARNING DOMAIN
  Course workspace → resource → assignment → submission → feedback
```

PLANNING DOMAIN

Goal → plan → task → deadline → progress

SUPPORT DOMAIN

Request → case → referral → appointment → resolution

FINANCE DOMAIN

Account → charge/credit → statement → payment allocation → reconciliation

CAREER DOMAIN

Evidence → permitted claim → portfolio → opportunity → application

GOVERNANCE DOMAIN

Policy → permission/approval decision → audit → assurance evidence

COMPANY DOMAIN

Lead → customer → agreement → implementation → subscription → renewal

Connections across these domains should reference canonical identifiers while preserving access scope, source authority, and lifecycle rules. Your mainframe document calls for canonical records plus source, freshness, classification, version, retention, and audit metadata. [1]

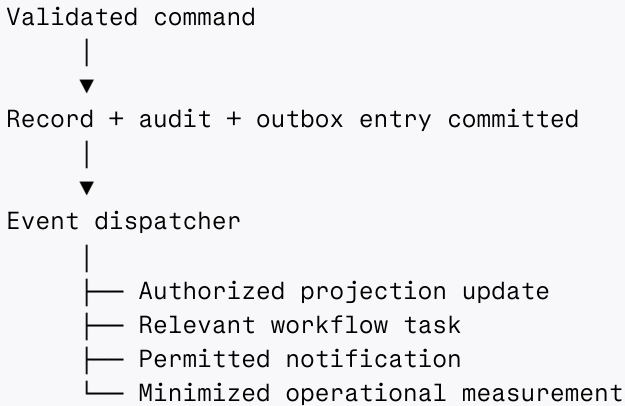
Proposed event network

The following names illustrate event contracts to define—not claims that these events currently exist:

Proposed event	Producing domain	Possible authorized consumers
membership.activated	Identity/membership	Onboarding, workspace routing
onboarding.first_value_achieved	Onboarding	Activation reporting, appropriate lifecycle messaging
enrollment.confirmed	Registration authority	Schedule, roster, course access
assignment.published	Learning	Student course view, notifications
grade.released	Authorized assessment workflow	Student grade view, approved record handoff
referral.accepted	Support/service workflow	Case status, responsible service queue
payment.settlement_confirmed	Verified finance adapter	Reconciliation, permitted account update
consent.revoked	Consent	Sharing enforcement, relevant cache/access invalidation
tenant.rollout_paused	Release control	Deployment operations, status workflow
membership.ended	Identity/lifecycle	Session/access revocation, ownership review

Each event contract should define its producer, schema version, tenant scope, minimal payload, access rules, retry behavior, and consumer responsibilities.

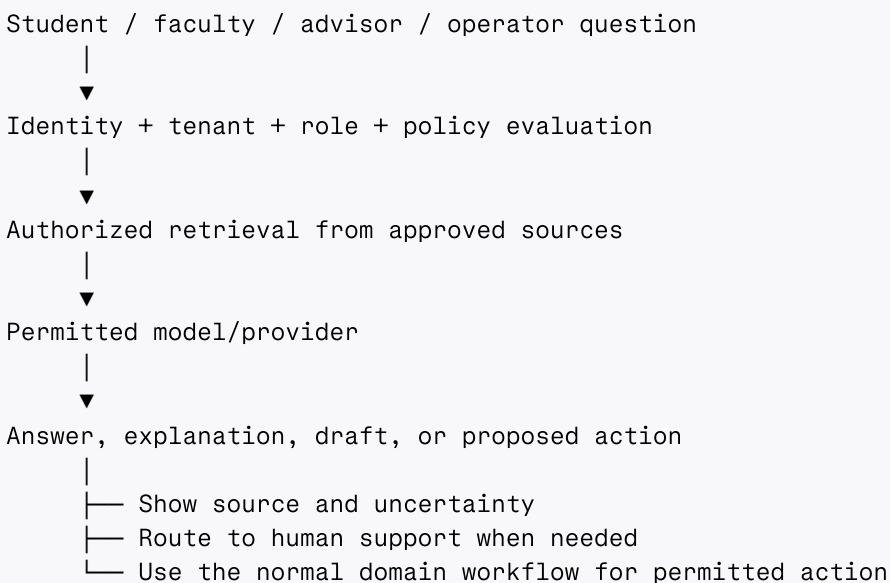
Events are not unrestricted broadcasts



For the proposed implementation, use restricted references rather than unnecessary sensitive content in event payloads. Consumers should handle duplicate delivery, stale events, failure, and reconciliation explicitly.

The shared command, audit, outbox, and projection approach is part of the supplied architecture. ^[1]

AI connects through the same boundaries



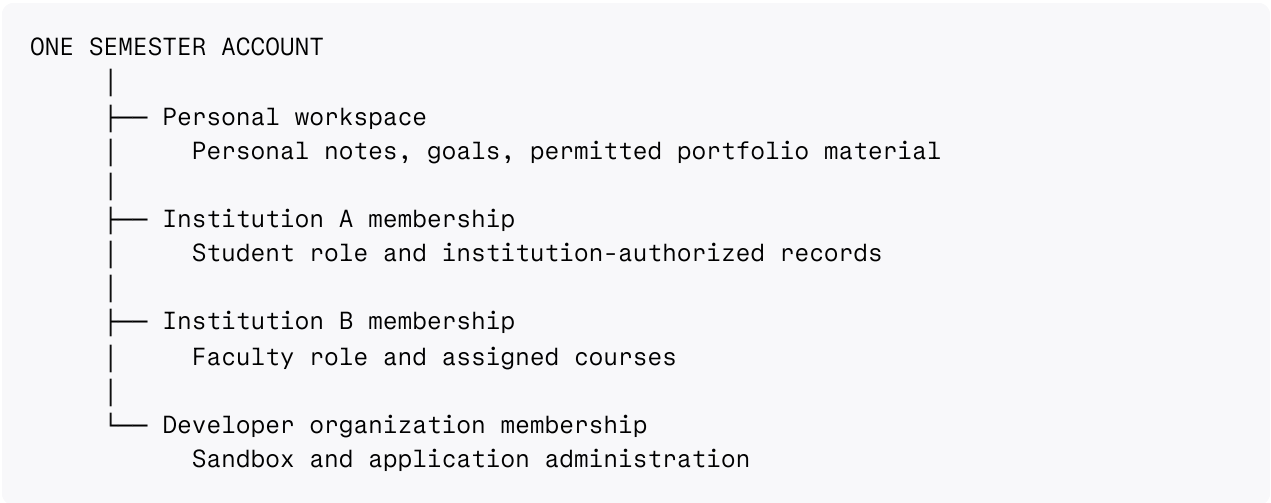
AI must not become a parallel authority. Your specification places official grades, academic-record changes, enrollment, financial approvals, and policy overrides behind authorized institutional workflows. ^[1]

5. Boundaries and ecosystem readiness

The expanded ecosystem needs clear limits as well as connections.

Multi-institution identity

A proposed person-level view should look like this:



One account helps continuity. It must not silently permit cross-institution access, merge official records, or transfer institution-held data into a personal workspace.

Likewise, leaving an institution should not automatically mean losing the personal account—but continued access to institution-held records must follow the applicable contract, authority, and retention rules. Your document distinguishes account continuity, role lifecycles, portability, and institutional record boundaries. ^[1]

Native ownership versus dependencies

Semester should own natively	A dependency may still supply
Company site, product messaging, conversion workflow	Hosting and content delivery
Signup experience, membership logic, workspace routing	Institutional SSO authentication
Course and academic workflows	Legacy authoritative records during transition
Checkout, invoices, account logic, reconciliation	Authorized payment processing and settlement
AI policy, retrieval permissions, evaluations, UX	Model inference
Notification rules, preferences, templates, history	Email and push transport
Developer scopes, installation approval, marketplace governance	Partner application functionality

Your mainframe document explicitly distinguishes replacement bridges from continuing identity, banking, compute, delivery, and ecosystem relationships. Native product ownership does not make all of them temporary. ^[1]

What makes a connection complete

For every proposed connection, require:

Who initiates it?
What outcome does it create?
Which domain owns the authoritative record?
Which roles may participate?
What information crosses the boundary?
Which policy or sharing basis applies?
Does it require approval?
What happens on failure or stale data?
What audit evidence is recorded?
Who supports it?
How does it end, revoke, reconcile, or migrate?

Readiness by connected journey

Instead of reporting “the dashboard is finished,” evaluate whole journeys:

Connected journey	Evidence of completion
Marketing → student activation	Correct CTA, account flow, resumed context, meaningful saved action
Institution identity → faculty course	Verified membership, assigned course, scoped teaching action
Student → advisor → service office	Proper referral routing, minimized context, resolution feedback
Registration → roster → schedule	Authoritative confirmation, consistent projections, error handling
Checkout → settlement → account	Verified payment state, reconciliation, accurate receipt
Customer contract → deployment	Approved scope, configured tenant, tested access, launch readiness
User issue → support → engineering	Scoped support context, usable telemetry, accountable resolution
Graduation → alumni continuity	Authorized credential action, explicit retained access, privacy controls
Partner app → institution install	Reviewed scopes, approved installation, monitoring and revocation
Role termination → access removal	Revoked access, transferred work, retention and audit evidence

Your readiness model already requires distinguishing design maturity, implementation maturity, deployment readiness, and record authority. The ecosystem expansion should apply that model to these connections—not only to individual components. ^[1]

The defining feature of Semester is therefore not the number of modules. It is the ability to move a real person’s work from one authorized workspace to another—preserving context, ownership, source authority, accountability, and continuity throughout the education and company lifecycle.

1. Show-and-break-down-entire-semester-mainframe.pdf